

SMALL WORLDS, INFINITE POSSIBILITIES? HOW SOCIAL NETWORKS AFFECT ENTREPRENEURIAL TEAM FORMATION AND SEARCH

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The social network perspective has become an important analytical lens for understanding strategic actions among entrepreneurs. Social theorists offer two competing visions of networks' configurations: one of infinite opportunities for individuals to develop heterogeneous circles of affiliations and the other of constrained opportunities privileging only certain individuals. We draw on this tension to describe three models of network formation – random, small world, and truncated scale free – and apply them to entrepreneurial team formation and resource mobilization strategies undertaken by entrepreneurs. We compare and contrast two models of team formation – a rational process model and an interpersonal relations model – and identify the network contexts under which each is most applicable. Mundane entrepreneurial teams arise within localized clusters and appear unlikely to take advantage of what network theorists have called small world networks, which depend upon bridging ties between clusters. Nonetheless, there are entrepreneurial strategies through which new ventures might achieve the advantages of small world networks. To the extent that new ventures emerge in truncated scale free networks, their founders must work within a highly centralized structure, with its institutionalized standards making team formation and entrepreneurial search more instrumental than within small worlds. Copyright © 2007 Strategic Management Society.

Early in the twentieth century, Georg Simmel (1955) published his classic essay on the intersection of social circles (translated into English as 'the web of group affiliations'). Contrasting social life in a metropolis with life in a village, he analyzed the

impact of modernity on individuals' chances of meeting dissimilar others and moving to new social positions. Joseph Schumpeter (2003), writing in the same era, argued that the old economic order was being transformed by ambitious people behaving entrepreneurially and improving their economic positions. For many other social theorists of that era, the early twentieth century was a time of turmoil and change in which 'traditional' ways were giving way to the 'modern,' a transformation that met with their approval.

By midcentury, the sense of optimism conveyed by early modernist writings had given way to more

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cautious sentiments. Two world wars and the onset of the cold war made theorists wary of promoting too optimistic a worldview. Publications on themes such as *The Organization Man* (Whyte, 1956) and *Small Town in Mass Society* (Vidich, 1958) suggested that individuals were heavily constrained by social structures, rather than agents of their own fate. Managers, rather than entrepreneurs, were the economic actors lionized by the news and entertainment media.

Over the past several decades, however, social theorists have once again returned to the issues raised by Simmel's analysis. One of the most highly cited papers ever published in sociology, Granovetter's (1973) 'The Strength of Weak Ties,' posed the question of the extent to which people's locations in social structures enabled them to escape the constraints of their social origins. Scholars who cite Granovetter's work often overlook one of his key themes, which was whether 'modernization' increased the likelihood that the job seeking and recruiting process operated on a universalistic, as opposed to a particularistic, basis. Like Simmel, Granovetter (1973) wondered whether modernization and structural differentiation emancipated people from ascriptive constraints. He studied employees, not entrepreneurs, but subsequent researchers saw value in extending his perspective to entrepreneurship.

In one of the first explicit calls for a social network perspective on entrepreneurship, Aldrich and Zimmer (1986) argued that putting entrepreneurial intentions into action requires that entrepreneurs establish connections with others who control resources. They argued that **social structures played a major – if not dominating – role in who tries to become an entrepreneur and who succeeds.** In this paper, we **build on their argument, adding** Simmel and Granovetter's **concerns for social inequality and achievement** to Aldrich and Zimmer's analysis. We argue that a theory of entrepreneurship and social networks must encompass the entire life cycle of new organizations, and that the best place to begin is with **understanding the role that social networks play in business startups: the formation of entrepreneurial teams and the mobilization of resources via social relations.** We present **three simple models of network structure: random networks, small world networks, and truncated scale free or 'fat tail' networks.** In describing each, we introduce concepts central to the literature on social network analysis, highlighting areas where we think network

analysis has something to say to entrepreneurship scholars.¹

WHAT DO NETWORKS DO?

Social network scholars in the early 21st century have clearly been inspired by Simmel's work of more than a century ago. In his essay on the web of group affiliations, Simmel (1955: 151) wrote, 'opportunities for individualization proliferate into infinity also because the same person can occupy positions of different rank in the various groups to which he belongs.' This passage on *infinite opportunities* appeared toward the end of the book, in a section on 'Individualism and Multiple Group Affiliations' in which Simmel contrasted the 'modern pattern' of social webs with the earlier pattern of 'concentric affiliations.' **Concentric affiliations cosseted people in successively larger rings of attachment, such as immediate family, extended kin, other village members, and the entire local community. The new web pattern gave people fresh choices and created opportunities for them to vary their positions across the various groups with which they affiliated. Individuals were no longer 'stuck' in fixed positions for their entire lives.** The business strategy literature has echoed Simmel's positive characterization of the potential inherent in social ties over the past several decades (Hoang and Antoncic, 2003).

Social networks in the startup process

The term 'network' is now firmly embedded in the business strategy literature, generally seen in a positive light. Most strategic network research focuses on network structures and their consequences within and between established firms. There are very few papers on the genesis of ties and even fewer that consider the role of networks in the founding of new ventures, with the exception of a few works

¹ We present the three models in sufficient detail so that they serve as baselines against which to analyze observed networks. In entrepreneurship research, model building is often the missing element in the theory-model-data trilogy of the scientific realism approach (McKelvey, 2002). In this approach, theory is used to inform the modeling process, which in turn informs analysts' research designs and data collection. Comparisons of research findings with models implied by theory allows us to improve the models and ultimately to build better theory.

such as Beckman, Burton, and O'Reilly's (2007) paper on the gains to new ventures from founding team members' career histories. In this paper, we extend social network concepts and principles to the early stages of a firm's life, focusing on the team formation and resource mobilization strategies in new ventures.

First, new ventures derive many instrumental and social-psychological benefits from teams (Aldrich, Carter, and Ruef, 2004; Birley and Stockley, 2004). Studies uniformly find that teams do better than solo founders, regardless of the measured outcomes. Second, when founders do not have direct access to the resources they need via team members and their ties, they often turn to indirect ties for help. For example, one group studying new ventures, noted that founders 'relied on pre-existing networks' as the primary means of access to the welter of resources needed during and after founding' (Baker, Miner, and Easley, 2003: 265). Specifically, we are interested in the way in which entrepreneurs can strategically search for resources using indirect ties.

Team formation

Two principles of team formation dominate the strategic management literature on teams, although most research concerns the upper echelons of established firms, rather than startups (Forbes *et al.*, 2006). First, a rational process model of team formation emphasizes selecting members based on pragmatic instrumental criteria, such as complementary skills or work experiences. From this viewpoint, competency should shape team formation so that new ventures possess the capabilities needed to manage complexity and growth. Second, a social psychological model emphasizes the interpersonal fit between team members and the need for smoothly functioning group processes. Many scholars have pointed to the important role that social and emotional support play in affecting human behavior (Thoits, 1984). For example, positive social relations within a team can create a supportive context within which people are encouraged to undertake innovative actions.

The two sets of principles are not mutually exclusive. Within the constraints of interpersonal attraction, teams can still search instrumentally for members. Similarly, within the constraints of resource-based needs, teams can still choose people who are 'attractive.' Nonetheless, as a normative principle, the business strategy literature seems to

privilege the rational process model over the social psychological model, claiming that new members ought to be chosen based on knowledge demands and resource connections, whereas accumulating evidence shows that 'new member addition, as it actually occurs, may be better explained by social-psychological theories' (Forbes *et al.*, 2006: 232). A social network view of team formation explains why such theories are a better fit to the realities of team formation.

We turn now to three models of network structure that are useful templates for understanding the social structural context within which teams form and entrepreneurs maneuver for opportunities and resources.

HOW DO SOCIAL NETWORKS ARISE?

Entrepreneurship scholars have embraced the optimistic message conveyed by much of the social network literature, viewing networking as the key to evading constraints on entrepreneurial action. Peter Blau (1977) gave scholars reasons for optimism, arguing that in modern society, people are able to participate in multiform heterogeneous circles and create many weak but integrative ties across diverse groups. Individuals thus gain numerous opportunities to experience diverse points of view. By implication, entrepreneurial opportunities are available to all.

In contrast, Pescosolido and Rubin (2000) argued that modern groups are so transitory and contingent that they do not really give people a basis for stable ties. Instead, people experience serial, ephemeral, short-term, contingent relations with others, mostly through indirect rather than face-to-face contacts. Pescosolido and Rubin cited growing temporary and part-time work as an example of increasing tenuousness and fragility in social relations. In describing the consequences of this postmodernist turn in the structure of contemporary social life, Pescosolido and Rubin (2000: 65) wrote, 'the expansion of roles that require flexibility, skill, and adaptability advantages some individuals while disadvantaging others. Those who profit from the new social forms are likely to be those in power (as in all social forms), those who are resource rich, and those on the cutting edge of societal developments (e.g., technology).' Note how they have taken Simmel's optimistic view of the transformation in the structure of

social relations, as well as Blau's image of society as integrated through cross cutting ties, and turned it into an expectation that the benefits from the new structures will fall unequally across social locations.

We use these musings on the changing social order to introduce our **three network models. For each model, we focus on two issues: (1) the extent to which the social world is organized into local clusters of densely connected individuals who interact primarily with one another, and (2) the average path length between individuals in the network, conceptualized as the average number of intermediaries it takes to connect any two randomly chosen individuals.** The first issue speaks to the issue of recruitment into entrepreneurial teams across social locations and the second sheds light on the dilemmas involved in entrepreneurial searches for opportunities and resources.

The first model we present, one of random networks, presumes a highly individualized world in which everyone has nearly unlimited access to everyone else, constrained only by limits on the resources that can be devoted to the search for new social ties. **Paths between distant people are short because there are no constraints on who can interact with whom** and so everyone is available as an intermediary or broker. **The second model, one of small world networks, presumes a highly clustered world in which people's searches for new ties are highly circumscribed by their environments,** with people's ties connecting them mainly to others in their same social context. Despite the potential for many failed connections between clusters, **small world theorists posit that bridging ties serve as short cuts connecting many of these local clusters to other clusters,** potentially reducing average path lengths to those found in random worlds. In *some* small worlds, the distribution of ties follows a power law and they are therefore an instance of our third model, the 'truncated scale free' network.

A scale free network model presumes that social networks arise through a process that results in a robust and highly structured hierarchical system that is highly resistant to disruptive events. Path lengths are short because a small number of highly connected nodes dominate the distribution, with many nodes having a small number of ties. These networks may also have local clusters, making them also small worlds. Empirical observation of social networks has found that few resemble pure scale free networks. Instead, the distribution of ties follows a power law

distribution with a 'fat tail,' and thus some analysts call them 'truncated scale free' networks.²

Model 1: Random Networks

Beginning with a random network seems counter-intuitive, as most people certainly do not think of themselves as behaving randomly as they go about their daily activities. How can we justify this choice? In fact, statistics based on random graphs are the baseline against which social network researchers judge observed graphs, as well as alternative models of social ties. By first considering a world without order or design, researchers are better able to appreciate the extent to which an observed network is truly special or unusual. Indeed, researchers follow the same principle whenever they conduct a statistical analysis of quantitative data. Analysts conduct tests of statistical significance against a baseline model of random association between variables, and when patterns of association depart significantly from randomness, search for reasons why. In the case of social networks, a random graph makes an interesting baseline because it exhibits a short average path length between nodes. Understanding why that is the case helps us appreciate the extent to which clustering, by itself, *reduces* global connectivity within nonrandom networks.

The structure of random networks

Two characteristics of random networks are central to our discussion. First, there is no clustering of nodes – nascent entrepreneurs in our case – and thus we have the most 'egalitarian' network in the sense that people's current locations do not limit their access to others. Neither prestige nor wealth affects someone's ability to connect to anyone else in the network. Second, average path length between nodes in random graphs is quite short because of indirect ties.

To convey a sense of the potentially unlimited connectivity in a world of randomly structured social networks, we offer a simple scenario. Consider a

²In truncated scale free networks, the degree distribution of ties is scale free over a significant portion of its range but it is not strictly scale free in the tails. In particular, at the 'high' end of the degree distribution, there are fewer extremely well-connected nodes that would be expected. Instead, the distribution bunches up short of the high values; hence the term, 'fat tail.'

situation in which an entrepreneur seeks resources from providers beyond his or her immediate set of direct ties – people known directly on a face-to-face basis. Assume that the entrepreneur ('ego') has 100 direct ties with other individuals ('alters') in his or her network. Then assume that each of the 100 alters has 100 direct ties in their networks. At this point, ego can access 10,000 additional individuals indirectly through the 100 alters with whom ego has a direct tie (i.e., $100 \times 100 = 10,000$ ties). If we assume each of the 100 first-order alters also has 100 direct ties with a second-order alter, ego can access an additional one million individuals indirectly (i.e., $100 \times 100 \times 100 = 1,000,000$ ties).³

Mathematically, we can formalize the logic underlying our example as follows. Define k as the number of ties from each person. Define N as the total number of people in the population under study. In a random network, the average number of steps needed to connect anyone to everyone else, defined as d , can be determined by the formula:

$$d = \ln N / \ln k$$

In the case of our example shown in Figure 1, $d = \ln 1,000,000 / \ln 100$, with d thus equal to 3. We can also solve for $N = k^d$. Simply put, in a random world, everyone is connected in a very small number of steps and so the average path length is very short. For example, in a world of 10 billion people, with $k = 100$, d is only five!⁴

To the extent that nascent entrepreneurs inhabit a world structured through random connections, two interesting implications follow. First, the lead founders of new ventures would face few social constraints on the recruiting of new team members. A random world would give them access to everyone and the potential of building a 'dream team' of founders. Second, although nascent entrepreneurs would face few constraints on their search, they would also find navigating in such a world highly problematic. The potential benefits of enacting new direct ties would be obscured by an entrepreneur's inability to predict

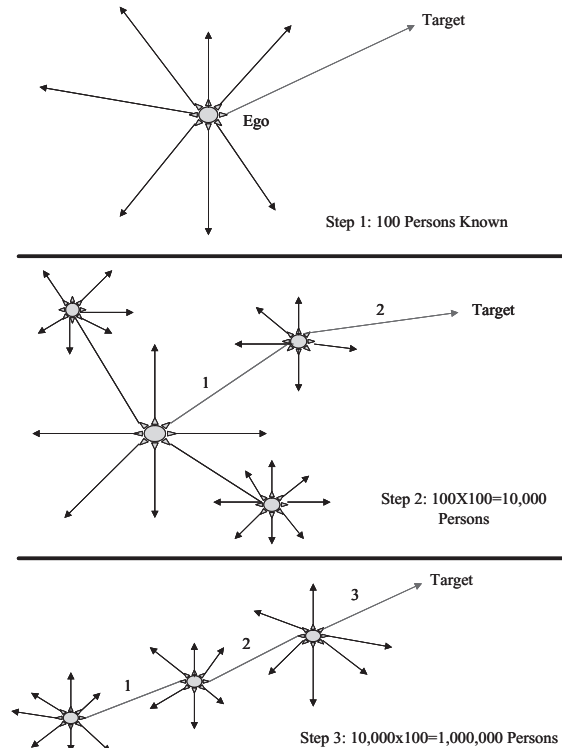


Figure 1. The potential of indirect ties

the payoff of indirect ties lying beyond them.⁵ We turn now to the much more structured networks of small worlds.

Model 2: Small World Networks

The 'small world' concept had been around for some time when Stanley Milgram began his ingenious series of experiments (Milgram, 1967). Milgram argued that even individuals who appear to have a constricted set of relationships could actually be connected to socially and geographically distant individuals. He derived the famous proposition that a person appeared able to reach any other person in just six steps. Subsequently, Watts and Strogatz (1998) identified the mathematical conditions under

³ As we note later, strictly speaking, these specific results obtain only if there is no overlap between direct ties, that is, at each step, ties are extended 'without replacement.'

⁴ Higher values of k decrease d , but at a decreasing rate. In our example of $N = 1,000,000$, when $k = 500$, $d = 2.2$, and when k increases to 1000, d drops only to 2. By contrast, for $k = 10$, d rises sharply to 6.

⁵ One reader of our paper wondered whether knowledgeable and skilled entrepreneurs would do better than others, even in a random world, because they would pick 'the best' direct ties. Unfortunately for these skilled entrepreneurs, knowing a direct tie's personal characteristics is of no help whatsoever in a random world, because they provide no information about that person's direct ties.

which it made sense to speak of a network as a ‘small world.’

Despite the enthusiasm with which the public and some academic researchers embraced the concept, many remain skeptical about the empirical incidence of small worlds. Kleinfeld (2002) became interested in the empirical basis for the six degrees of separation principle and so looked into the archives of Milgram’s studies. She found that the vast majority of the chains begun at originating sites did not reach their destination sites. She searched for other studies that had replicated the small world study and found only one that spanned at least two disconnected cities. Its completion rate was also very low. Moreover, that study underscored a substantial racial divide in the extent to which connections were completed. Kleinfeld’s skepticism encourages further assessment of the assumptions underlying the ‘small world’ phenomena. Thus, in this section, we will examine those assumptions and what they might tell us about the conditions under which nascent entrepreneurs form startup teams and search for resources.

The structure of small world networks

Models of small world networks differ from models of random networks in two important ways (Watts, 1999). First, rather than connections being formed randomly across entire populations, relationships are clustered together in local networks, such as neighborhoods, friendship circles, or workplaces. Such clusters form because socio-cultural constraints substantially limit the extent to which any two persons might encounter one another. Most ties are based on homophily, rather than randomness (Ruef, Aldrich,

and Carter, 2003). These conditions increase the density of connection within clusters and reduce the chances that people in one cluster will have contacts with people in other clusters. By making it more difficult to enact ties with dissimilar others, such clusters potentially raise the average path length in social networks.

Second, to meet the condition of low average path lengths within small worlds, theorists posit that local clusters of nodes are linked to other local clusters through bridging ties. These bridging ties link clusters together to form a global network. Small world theorists have shown that a surprisingly *small* number of long-distance bridging ties have to be added to otherwise fragmented social networks to create a small world (Watts, 1999). The bridging ties jump over otherwise broad gaps in the network, thus lowering the average path between any two points. Note that small world theorists do not make claims about the *frequency* with which small world networks occur in the real world, but only that networks meeting these two conditions will be small world networks. Whether any particular network meets the conditions is an empirical question.

In Figure 2, we display two networks. For both networks, we calculate the clustering coefficient (based on the number of complete triads divided by the total number of potential triads where at least two out of the three nodes are directly linked) and the average path length (based on the number of links between all pairs of two nodes that could be reached from one another and the total number of linked nodes). The network in panel A does not have any nontrivial clusters and lacks bridging ties across distant nodes. In contrast, the network in Panel B

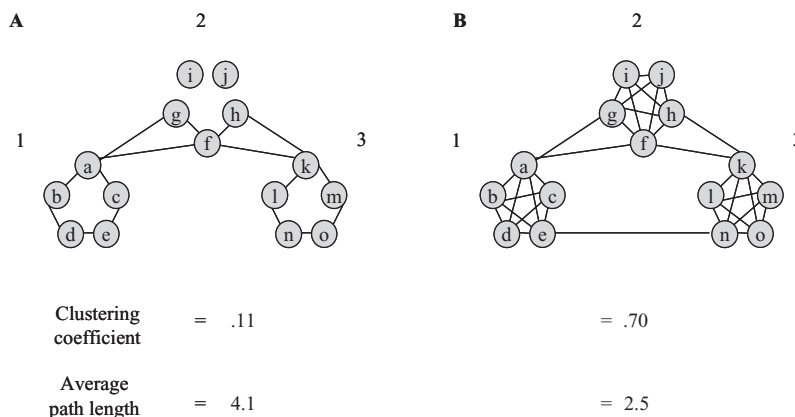


Figure 2. Small world examples

contains several small world clusters and bridging ties joining these clusters. As a result, the clustering coefficient increases significantly between the two networks. Clustering coefficient values can range from a low of zero to high of one, and given the values in the figure, it is clear that a much higher percentage of nodes in panel B are connected to each other. Additionally, the average path length decreases substantially with the addition of a few bridging ties. In principle, people are much closer together in Panel B than in Panel A.

Homophily and clustering

Social science research strongly supports the proposition that social structural conditions generate densely linked local clusters. Researchers have extensively documented the generalization that ‘birds of a feather flock together’ since the early 20th century. Studies range from research on friendships (Lazarsfeld and Merton, 1954) and teams (Ruef *et al.*, 2003) to studies of cultural and voluntary associations (Emerson and Smith, 2000; McPherson and Smith-Lovin, 1986) and business organizations (Ibarra, 1995; Kanter, 1977). *Homophily* constitutes the central principle behind these consistent findings: ‘Homophily is the principle that a contact between similar people occurs at a higher rate than among dissimilar people’ (McPherson, Smith-Lovin, and Cook, 2001: 416). In the language of social networks, the ‘friends of our friends’ are already our friends, rather than strangers unknown to us.

Local clusters form in ways that sustain and amplify homophily. Strong boundaries deflect social relationships back upon themselves, thus fostering highly concentrated social networks. For example, instead of extending an open public invitation to join, existing clusters recruit new members either by specifically recruiting them (e.g., LinkedIn users send an invitation via email to other potential users) or through drawing on their current ties in other domains (e.g., through kinship ties). Because individuals who share similar characteristics are more likely to know each other, these individuals tend to form dense clusters in which everyone knows everyone else.

Dense local networks can lead to three self-reinforcing dynamics that reproduce and magnify their tendency toward narrow clustering. First, new people wishing to join an existing local network face potential barriers when they do not share common experiences and interests with current members. For example, kin-based network clusters may squeeze out

non-kin who would otherwise benefit from membership (Marsden, 1987). Second, similarities among individuals create stronger linkages and reduce the likelihood of turnover within a local network. Conversely, dissimilarities increase the likelihood of people leaving relationships. Third, if homophily serves as a basis for recruiting similar others, this common characteristic could be used as a screening mechanism. These three self-reinforcing dynamics, driven by homophily and propinquity, sustain the homophilous composition of local networks.

Bridges and short cuts

Because of clustering, people in discrete parts of a network will be unlikely to encounter people in other clusters without going to extraordinary lengths, such as working through many intermediaries. However, small world theorists have noted that it does not take many bridges between clusters to lower substantially the distance between people in different clusters. For our purposes in studying the creation of new ventures, we need to understand the process by which such bridges could be constructed. What strategies might entrepreneurs employ to shorten paths to potential opportunities and resource providers?

Constructing bridges across clusters poses a major challenge to most people. Although most of these boundaries are quasi-permeable, surmounting them requires work that people are often discouraged from undertaking. People’s peer groups often actively discourage contact with dissimilar others, and settlement patterns create geographic separation that individuals must work to overcome.⁶ Restrictions on associative activities create a recursive cycle – given knowledge constraints, individuals become habituated to seeking out similar others and uncomfortable with dissimilar others (McPherson and Smith-Lovin, 1987).

Overcoming these constraints requires that individuals proactively make strategic choices that push

⁶For example, Mouw and Entwisle (2006) used social network and spatial data from the National Longitudinal Study of Adolescent Health (Add Health) to examine the effect of racial residential segregation on school friendship segregation in the United States. They attributed about one third of the level of racial friendship segregation in schools to residential segregation, indicating that where students lived had a powerful effect on the likelihood that they were able to form cross-racial social ties. More of the effect resulted from residential segregation across schools than within them, implying that many students were not in a position to form inter-racial friendships because they had little chance of meeting people who were not like them. In short, they had few out-group choices to make.

them across social boundaries, or that individuals become involved in activities that expose them to dissimilar others. Individuals who pursue this strategy must be prepared to bear the additional costs generated by bridging differences (Arrow, McGrath, and Berdahl, 2000; Popielarz and McPherson, 1995). For example, Davis, Renzulli, and Aldrich (2006) found that nascent entrepreneurs who wished to increase the diversity of their social networks derived few benefits from simply joining a diverse set of voluntary associations. Instead, they gained network diversity by actively participating in the associations and aggressively seeking out new contacts at meetings. As another example, commercially oriented social network websites formalize the process of searching for bridging ties by showing members that a bridge exists between them and a desired contact whom they do not know personally. It is then up to them to convince the intermediary, whom they know, to make introductions for them.

Summary

The principle of small world networks reveals the tension hidden in calls for nascent entrepreneurs to 'use networking' to build entrepreneurial teams and seek opportunities. Being embedded in dense clusters of social relations that have emerged through homophily and propinquity creates a highly circumscribed world for entrepreneurs. However, to the extent that bridging ties allow entrepreneurs access to distant regions of a global network, the potential scope for entrepreneurial action expands enormously. Such ties can be an emergent consequence of forces beyond their control, but entrepreneurs can also reduce path lengths to valuable resources by taking strategic action. We will return to this point in a later section.

Model 3: Truncated Scale Free (or 'Fat Tail Distribution') Networks

Our third network model, of truncated scale free networks, differs in several ways from random and small world networks. Small world networks can also be scale free, whereas by definition, random networks are not. Thus, we begin by highlighting the differences between random and scale free networks and then discuss their relation to small world networks. As we noted earlier, few networks studied by social scientists are purely scale free but rather

follow an exponential distribution with a 'fat tail.' For ease of exposition, we will keep the term 'scale free' as we describe the pure model.

Scale free compared to random networks

Scale free networks follow a power law in the distribution of their nodes and ties, unlike random networks, which follow a Poisson distribution. In scale free networks, some nodes have a very large number of ties (in and out) and most of the rest have very few. By contrast, random networks have a peaked distribution in the number of ties per node, and the distribution is very homogeneous, centered on the mean and mode. We illustrate the differences using the graphs displayed in Figure 3.

Panel A of Figure 3 shows a random graph with 36 nodes and 44 links, with an average in-degree of 2.42 ties per node and a very small standard deviation of 0.81 (Kurakin, 2007). Based on the figure in Panel A, Panel C shows the distribution of ties, $P(k)$, which is the probability that a randomly chosen node in a network has exactly k links. So, on the y axis, $P(k)$ is the fraction of nodes that have k links. Because Panel A is a random network, $P(k)$ in Panel C follows a Poisson distribution, which peaks at average k and exponentially decays as k increases, at rate $P(k) \sim e^{-k}$. The network in Panel A is not very highly interconnected, as only 44 out of 630 possible links are present. Using the formula we presented earlier in describing random graphs, the average path length between nodes would be about four.

Panel B shows a scale free network, also with 36 nodes and 44 links, with the same average in-degree of 2.42 as in the random network, but with a larger standard deviation of 2.63. However, in this case, the *average* is extremely misleading as an indicator of connectivity. As in all scale free networks, a small number of hubs dominates the distribution, with the top three having 9, 10, and 11 links to them, respectively. Panel D graphs the distribution of the figure shown in Panel B, and shows that $P(k)$ follows a power law distribution. There is no single peak, and the log/log plot in Panel D shows the $P(k)$ distribution follows the formula $P(k) \sim k^{-\gamma}$. Barabási (2002) noted that most of the power law distributions he studied had a γ of between 2 and 3. Thus, even though both networks have the same number of nodes and links, they are quite different in their organization. In the random network, the contribution of any particular node to the connectedness of the overall network is roughly equal, whereas in the

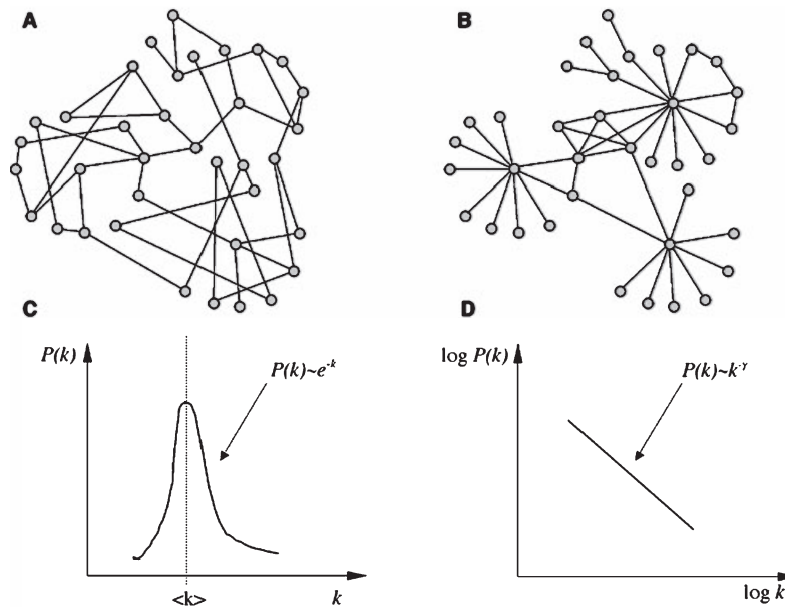


Figure 3. Examples of random and scale free networks

Note: Panel C graphs the degree distribution shown in Panel A. Similarly, Panel D graphs the degree distribution shown in Panel B.

Source: Source: <http://www.alexekurakin.org/>
Accessed June 26, 2007. Used with permission.

scale free network, a small number of nodes dominate the structure.⁷

Scale free compared to small world networks

Small world networks can also have a scale free distribution of ties. Unlike the small world networks we described in the previous section, the connectivity of a scale free network does not depend on shortcuts closing long-distance gaps between clusters. As an example, consider again the small world network shown in Panel B of Figure 2. It has a much shorter average path length than the network in Panel A, but has a more uniform degree distribution than that of Panel A. In Panel A, the average node has 2.13

ties and the standard deviation for the distribution is 1.18, whereas the average node in Panel B has 4.73 ties and the standard deviation is only 0.88. The reduction in path length between the two networks was not achieved through a small number of highly connected nodes, as in Panel B of Figure 3, but rather through a few links serving as short cuts between clusters.

How might scale free networks emerge?

Barabási (2002) suggested that scale free networks are best viewed dynamically, not statically. He and his collaborators proposed two simple principles by which to understand how scale free networks grow: growth takes place one node at a time and new nodes link to existing nodes following a principle of preferential attachment. As each new node is added to the network, it links to a few others. These links are not made randomly, but rather the node ‘observes’ which nodes already have the most links and then link to them. For example, investigators studying vicarious learning have noted that firms can bypass the need for trial and error learning by choosing a simple rule of imitation, such as ‘imitate the apparently most

⁷Not every highly centralized network has a scale free degree distribution. In designed networks, such as those inside large firms, there is an intra-organizational hierarchy, with a chief executive officer at the top and then descending groups below, in a ladder like structure. Thus, there is a uniform degree distribution by design, following guidelines about the number of direct reports to any particular manager, and the distribution of nodes does not conform to a power law. However, this is not a generic result for centralized networks. It cannot be extended to naturally emerging graphs in open networks.

successful individual or firm,' or 'imitate the fastest growing firm' (Miner and Haunschild, 1995). Thus, if entrepreneurs perceive that successful firms are funded by a top-tier Silicon Valley venture capital firm, such a rule would lead other entrepreneurs to seek investment with the same VC.

Following the principle of preferential attachment means, 'the rich get richer,' thus increasing inequality in a network. However, it is only probabilistic: not every new node links only to the most preferred. For growth to result in a scale free network, it is only necessary that a high proportion follow the preference principle. Some newcomers will *not* flock to the most highly connected nodes, because they have divergent preferences or because of capacity constraints on the ability of the most highly linked nodes to handle large numbers of ties arriving in a short interval. Nonetheless, if a large enough proportion of newcomers follow the principle of preferential attachment, the resulting network evolves into one with a few highly linked and many sparsely linked nodes.

What evidence do we have for the principle of preferential attachment driving network growth in the social world? We offer three examples. First, some ethnic business owners, in some communities, follow the principle of preferring to deal with other co-ethnics rather than outsiders (Aldrich and Waldinger, 1990). Second, studies of social movements and the 'bandwagon' effect, especially when driven by the mass media, show strong evidence for advantages accruing to pioneers (Low and Abrahamson, 1997). The race by dot.com firms to capture 'eyeballs' was premised on the existence of the preferential attachment principle, and studies have found that the structure of links to Web sites follows a power law distribution (Barabási, 2002).

Third, in fields where status and reputation effects strongly govern choices of which ties to form, we would expect scale free networks to form (Pollock, 2004; Pollock, Porac, and Wade, 2004; Stuart, Hoang, and Hybels, 1999). The venture capital industry represents a particularly promising place to look for scale free networks, based on status and reputation effects, given the high degree of uncertainty facing its participants. In their analysis of U.S. venture capital firms' investments, Sorenson and Stuart (2001) noted that frequent reliance on investment syndication, rather than solo investing, created a dense VC interfirm network that structured the flow of information in the industry.

Sorenson and Stuart's description of the network's evolution seemed to describe a preferential attachment pattern in which each syndication participation increased the likelihood of being invited again, as well as increasing trust in the syndication partners, thus raising the likelihood of future cooperation. New firms joining the network thus were at a disadvantage in linking with the most connected firms, in contrast to those already in the network. This pattern was confirmed in Kogut, Urso, and Walker's (2007) research on the development of the U.S. venture capital industry from 1960 to 2005, which revealed a more complex pattern of ties than a simple power law distribution. Rather than simply affiliating with the most connected firms, firms preferred to do new deals with incumbent firms that they had dealt with in the past. Thus, although the degree distribution itself deviated from a simple power law, the distribution of *repeated* ties followed a power law distribution. They suggested that the forces generating a nationally integrated venture capital network were more complex than simple preferential attachment and called for more research on the process (Kogut *et al.*, 2007).

Scale free networks and vulnerability to disruption

Scale free networks differ radically from random networks, and from small world networks that are *not* scale free, with regard to their vulnerability to failure. Random and non scale free small world networks are highly vulnerable to disruption, because dropping or cutting nodes randomly can disconnect the entire network, resulting in isolated fragments. In contrast, scale free networks are not vulnerable to failures at random. Many nodes can be removed randomly without disconnecting the overall network and breaking it into fragments. Most scale free networks studied by Barabási (2002: 115) and his students – who were explicitly looking at the issue of vulnerability – had a degree exponent smaller than 3 (the γ parameter in $P(k) \sim k^{-\gamma}$): 'Therefore, these networks break apart only after all nodes have been removed – or, for all practical purposes, never.' However, scale free networks are highly vulnerable to attacks directed at the most connected links – the top of the hierarchy, which has the nodes with the highest connectivity. As the most connected nodes are removed, the network eventually falls apart.

We can apply this idea to entrepreneurship in 'failed states' or in communities experiencing social upheaval. When a society or community experiences a major social upheaval, such as a civil war

or a natural disaster, purely random failures will not threaten its overall stability. However, if civil war or terrorism results in the killing or emigration of top leaders, or natural disaster drives out the most well connected members of the community, the social fabric is threatened. Coordination between social entities will be much more difficult and path distances between nodes will increase substantially. In the United States, New Orleans seems to be experiencing this type of failure, as many business owners have left and entrepreneurs have created few new ventures. Thus, high growth oriented firms will be less likely to start up and less able to flourish in these troubled environments. However, very small-scale entrepreneurship will be much less affected and may even be more likely to thrive with so many dominant players removed or neutralized.

We can speculate that entrepreneurship will also be difficult, if, for some reason, a social network cannot grow scale free. In addition to the social upheaval already discussed, other forces may hamper scale free growth. If cultural or social constraints prevent ordinary nodes from becoming well-connected hubs, then the network will be more random than scale free. For example, if a society's or community's norms strongly encourage equality and the egalitarian distribution of resources, then 'leveling' will affect the nodes that began to accumulate resources and the network will remain fragmented or more small-world like than scale free.

Summary

Pure scale free networks display a degree distribution that follows a power law, as we showed in Figure 3. However, many degree distributions found among social networks are better characterized as exponential with heavy or 'fat' tails. That is, their degree distributions look very much like the line drawn in panel D of Figure 3, except that rather than continuing in a straight line down to the x-axis, the distribution is truncated and curves to the x-axis.⁸ Commentators have offered various speculations on why this occurs, such as Kogut *et al.*'s (2007) observation about the tendency for repeated ties with the same players in the venture capital industry, rather than ties to the most linked players. For our purposes, we simply note that the truncated scale free

model is a good beginning point for understanding why average path length is so short in some kinds of networks.

APPLICATION TO ENTREPRENEURIAL TEAMS AND SEARCH

We turn now to two applications of the three network models we have reviewed. The first concerns the emergence of entrepreneurial teams and the consequences of network structures for team composition. The second concerns how network structures affect entrepreneurial search.

Entrepreneurship: Who's on YOUR team?

We noted earlier that the literature on entrepreneurial teams has proposed two principles by which teams might form. The rational process model of team formation emphasizes selecting members based on pragmatic instrumental criteria, whereas the social-psychological model emphasizes the interpersonal fit between team members. We argue that the instrumental model fits best those teams that emerge in well-institutionalized fields characterized by networks whose degree distributions resemble truncated scale free or exponential distributions with fat tails. By contrast, we believe the social-psychological model fits best for the vast majority of team foundings, in part because they are embedded in disconnected local clusters that only sporadically coalesce into small worlds that contain shortcuts to diverse other clusters.

Teams in well-institutionalized scale free networks

We can gain insight into the special social structural conditions under which most entrepreneurial teams form by examining teams emerging within networks characterized by power law distributions with fat tails: creative teams of coauthors in academic disciplines and creative teams in the Broadway music industry (BMI). In their work on team assembly mechanisms in four academic fields and the BMI, Guimerà *et al.* (2005) found that the set of people from whom teams were drawn was embedded in a larger network which acted as a storehouse of past knowledge created within the field. The large network of participants shared common professional standards, norms of collaboration, and was national in scope. Over time, the fields became increasingly

⁸They have fewer extremely well connected nodes than would be predicted by the exponent of the equation describing the log/log plot.

integrated, moving from semi-isolated small clusters to a single large connected cluster. A small number of very prestigious actors acted as brokers and created bridges between clusters, lowering the average path length for the entire network. Founders assembling teams thus had a very large community of practitioners from which to draw, with shared social identities.

Team sizes in all fields increased over time, reflecting the growing complexity of the fields and external performance pressures. Clearly, competence was a very important criterion in assembling teams and we would expect universalistic norms to govern recruitment. These teams exemplified many of the desired characteristics identified by the 'rational process' models of entrepreneurial team formation: adequate size, skill diversity, shared prior experiences, and high human capital. Even so, the investigators found a very strong tendency for people to repeat past collaborations, just as in the venture capital industry. At least for the BMI, the shape of the degree distribution could be characterized by a power law with an exponential tail, meaning it was highly skewed with a few people at the top with many ties and many people with only a few ties (Jarrett Spiro, personal communication). In this respect, it resembled the VC industry repeated tie distribution that we have already described (Kogut *et al.*, 2007), indicating that a process of preferential attachment seemed to be driving the national network for the BMI.

The social structural conditions underlying team emergence in these five fields provide a sharp contrast to the much more locally oriented networks out of which most entrepreneurial teams emerge. Nonetheless, they could well be descriptive of particular subfields of entrepreneurship, such as within regional clusters around Route 128 or Silicon Valley or within particular industries drawing on national talent pools where prestige and status affect the recruiting process.

Mundane entrepreneurial teams

In contrast to entrepreneurial teams formed in well-institutionalized fields, teams emerging under typical circumstances are deeply embedded in local clusters of social relations. We cannot assume that most nascent entrepreneurs are operating in a small world context in which ties to other clusters help them recruit diverse members for their founding teams. Whereas Guimerá *et al.* (2005) could take for granted an instrumental basis for team formation

as scientists searched for coauthors and Broadway producers sought choreographers and composers, we must acknowledge the priority of interpersonal relations in mundane foundings. Under such conditions, few teams will recruit out-of-cluster members. In this section, we review some evidence showing the extent to which homophily, familiarity, and proximity dominate the dynamics of team formation, supporting the interpersonal relations model.

Interpersonal forces within local clusters powerfully mold team formation because they create conditions of high trust and positive social support. First, according to Coleman (1988), densely packed networks of social relations within a cluster constitute a primary source of 'social capital' because social norms are monitored and enforced more easily within closed networks. In dense networks, when people violate norms, they suffer the consequences of local sanctioning, such as loss of reputation. Within closed networks, violators will confront a 'united front' composed of members who call for a remedy. For example, nascent entrepreneurs in many developing nations use rotating credit associations to raise the capital they need. They rely on collective trust among their members and take advantage of the benefits of network closure (Biggart, 2001).

Second, local clusters in which a high proportion of alters know one another can be useful because they provide social support and facilitate the transmission of complicated information (Nahapiet and Ghoshal, 1998). Such networks, also called 'high density networks,' can foster economic relations through the creation of trust. For example, Portes and Sensenbrenner (1993) argued that social closure within immigrant communities gives individuals greater access to financial resources. Immigrant entrepreneurs are thus able to finance projects via ethnic-based cohesive ties.

Homophily and familiarity in team formation

Using data from the Panel Study of Entrepreneurial Dynamics, a representative national survey of 830 people who reported they were in the process of trying to start a new business in 1999–2000, Ruef *et al.* (2003) showed that two principles dominated team formation: homophily and familiarity. As we noted, in reviewing the forces generating clusters in small world networks, the homophily principle refers, in general, to a tendency for people with similar characteristics to associate with one another. In the PSED, teams were extremely homogeneous

with respect to gender, race, and occupation. For example, racially homogeneous teams appeared in the data at a level 27 times that expected based on random mixing, and gender homogeneous teams – net of spousal pairs – were about 5 times more likely than expected. As we followed the founding efforts over time, we observed that they became more gender and racially homogenous as people left and new members joined the teams.

Familiarity is also a key social mechanism of in-group formation, and can be a result of pre-existing ties, such as through work or family. The principle of familiarity asserts that people who associate with one another, under certain conditions, become more likely to continue the association subsequently in other circumstances. How far outside their immediate circle are founders prepared to go in building a team? Research shows that people rarely establish ‘relationships’ with those they meet by chance (Grossetti, 2005). Instead, local clusters of family, friends, work, and neighborhoods will serve as the pool of people available for recruitment into entrepreneurial teams, if nascent entrepreneurs follow the principle of interpersonal relations in team building.

Following Simmel, we think it is useful to distinguish three concentric circles of inclusion for teams: family, others known personally to the lead founder or core founders, and strangers. Family members, particularly spouses and domestic partners, fulfill many of the requisites of shared identity that are otherwise generated by homophily. They are often in close proximity to one another and have had many opportunities to share rewarding experiences. Because they are likely to interact frequently, family members have many opportunities to discuss the possibility of starting a business together. They also have many occasions during which they can observe the reliability and trustworthiness of one another, perhaps predisposing them to a favorable response when asked to join in a new venture. If they see one another regularly, the chances for an escalation of commitment increase. Ideas that might be superficially discussed and dismissed in other contexts might lead, among kin, to the cumulative development of a plan for action.

Ruef *et al.* (2003) found that spousal and kinship ties were the most common basis of within-team relations: 65 percent of all teams had spouse/partner ties, other family ties, or both. Business ties seem to be an alternative to spouse/partner ties for team building, and about 39 percent of the non-spouse/partner teams had at least one such tie among its members. In

special analyses conducted for this paper, we found that only 4 percent of the teams organized around spouse/partner pairs also contained members linked by business or work relations. Larger teams were more likely to have such ties than two person teams.

To learn more about the way in which business ties influence team formation, we examined how the tie occurred. For the 173 ties reported in all 86 teams that had at least one business tie within the team, including eight spouse/partner based teams, we found that 51 percent were based on colleagues who worked at the same place and 49 percent were not. The process of workplace recruiting exemplifies the self-reinforcing pattern of homophily in social relations we mentioned earlier. In their work on hiring practices at a phone center, Fernandez, Castilla, and Moore (2000) reported that new hires referred by current employees were more likely to be similar to current employees than non-referred new hires.

We suspect that many of the ties not based on currently working together might reflect prior joint work experience in another workplace. Some of these ties might also reflect other kinds of business relations, such as clients, vendors, or consultants. Friendship and acquaintance ties were slightly more prevalent among the non spouse/partner teams than were business ties. About 44 percent of the non-spouse/partner teams had at least one pair of members who were friends or acquaintances prior to joining the team. By contrast, only 7 percent of the spouse/partner based teams also included at least one friends/acquaintance tie. Founders also call on others for help with specific problems, and our research shows that over 70 percent of founders have non-owner ‘helpers’ in their founding groups (Kim, Longest, and Aldrich, 2007).

Strangers need not apply

Strangers – people not related by kinship or known to one another prior to the initial interactions around the founding of a new venture – constitute the most interesting potential team members. Rational process theories of entrepreneurial team functioning strongly imply that interpersonal considerations are secondary to instrumental ones (Barney *et al.*, 1996; Mosakowski, 1998). If industry experience and technical or managerial competence are critical to the success of a team, then we would expect lead entrepreneurs to search widely for qualified people, using existing bridges to go beyond local clusters or creating new bridges if none exist. Well-qualified

strangers might even be preferred over less-qualified family, friends, and business associates.

Accordingly, perhaps the most striking finding from the PSED concerns the nearly complete absence of any strangers whatsoever among two person teams, and their rarity among three person teams. They were also extremely rare among all spouse/partner based teams, with only 2 percent of such teams reporting any stranger ties. Only among the non spouse/partner based teams with 4 or 5 members, representing 18 percent of all the non spouse/partner teams, did we find a sizeable proportion of strangers. About half of the three and four person teams had at least one pair of strangers, and almost three-quarters of the five person teams did. Nonetheless, even in these large teams, most team members still knew each other prior to team formation. Clearly, these strangers carry the possibility of serving as bridging ties to other clusters. We suspect that these large teams, somewhat dependent upon people who were unknown to one another before forming the team, represent the kinds of more capital-intensive and growth-oriented teams that figure so prominently in the strategic literature on entrepreneurial teams.

Several studies of new venture performance suggest that pre-existing relationships might be an advantage within a team, rather than a source of concern (Francis and Sandberg, 2000). Forbes et al.' (2006) study of academic spinouts from two large public universities found new team members were selected for their similarity to existing team members, apparently driven by a desire for team consensus and harmony. Thus, local embeddedness in a cluster may not be a disadvantage, because the benefits of positive interpersonal dynamics offset the losses from a more homogenous team.

Summary

Clearly, the great majority of entrepreneurial teams emerge out of the local clusters described by small world networks but *without* the bridging ties necessary to reduce the social distance to strangers qualified for team membership. As such, we would expect them to be very stable, and follow-up studies over the subsequent three waves of the PSED bore out that expectation (Kim and Aldrich, 2004). Indeed, only 12 percent of the new ventures identified in the first wave experienced any changes in team composition over the four waves.

Unlike the creative teams assembled in increasingly institutionalized fields in the United States, as

documented by Guimerá *et al.* (2005), most attempts at founding new ventures in the United States do not draw their members from a nationally-integrated pool of experts whose competence has been judged against agreed-upon standards. Strangers are rare, except for the largest teams, possibly hinting at a 'competency discount' that founders extend to potential members whom they know and trust. Instead, almost all startups assemble teams based on embedded ties from pre-existing relations within local clusters. We speculate – and we need more research on this issue – that founding team members use instrumental criteria mainly *within* networks of embedded ties. Apparently, bridging ties usually fail to bring in non-homophilous members. Most new ventures do not inhabit a 'small world.' To pursue this issue further, we turn now to the issue of bridging ties and entrepreneurial search more generally.

Search: Holed up in clusters?

At any given moment, the composition of entrepreneurial teams shows the impact of their embeddedness in local clusters and the constraints they face in breaking out of such clusters. However, networks are best viewed dynamically, not statically. If nascent entrepreneurs and founding team members find themselves closed off in clusters without indirect ties to the resources and opportunities they need, they can actively engage in resource mobilization through network-based search. Indeed, 'small worlds' depend upon bridges constructed through ties between clusters. In this section, we consider four questions: how does someone's location in a cluster affect their search behavior, how do people break out of clusters, what role does someone's social identity play in search, and how is search different in scale free networks?

Dense clusters inhibit search

High-density networks can be detrimental to the extent that they induce conformity and constrain individuals' autonomy, creativity, and innovation (Borgatti, 1997; Gargiulo and Benassi, 2000; Hansen, 1999; Putnam, 2000). When networks are dense, the potential assistance available from different members is often redundant because information and potential contacts are likely to be widely shared (Glanville, 2004). Additional network members in a dense network fail to provide novel information, although they might be able to supply other resources that entrepreneurs need.

Burt (2005) argued that people often provide others with information that they believe the person would like to hear, thus merely 'echoing' what the person already knows or believes. 'Echo' gives people a distorted view of what is really happening in their world, because it is redundant information, perhaps even deceptive. When coupled with people's failings with regard to self insight (Dunning, 2005), 'echo' falls well short of providing an accurate and reliable view of someone's environment. Social networks thus can affect the *meaning* that people find in their environments, by shaping and altering the information they receive and perhaps limiting their awareness of the need to search more widely for accurate information.

Breaking out of clusters

Consider a nascent entrepreneur contemplating recruiting a team of founders. Very few will methodically analyze their entire personal network, taking into account their weaker direct and potential indirect ties. Lacking clairvoyance, people will inevitably concentrate their investigations on their immediate circle of friends and any inferences about the value of new ties will depend heavily on personal intuition and experience. However, we note that networking software, such as LinkedIn, allows individuals to approach network searching in a much more systematic manner, and using such software seems on its way to becoming a 'best practice' in the high-technology sector.

Finding valuable information without the aid of sophisticated software may depend, in part, on a person's position in a social network. Indeed, Burt (2005) has forcefully argued that in explaining successful outcomes, investigators have given too much weight to human capital and not enough to location in social networks. His well-known argument about brokerage and closure stresses that advantages flow to people who recognize or at least take advantage of structural holes in networks. Throughout his analysis, he recognizes that people in 'good' locations might be different from others, and thus he always controls for the effects of human and social capital, e.g., age, experience, and job.

Implicit in his analysis is the assumption that people will recognize when they occupy bad locations and thus move to better locations. He seems to posit that, in the cross-section, we observe the consequences of people taking advantage of 'the kind of person he or she is, so motivation does not have to be measured once one has a measure of network structure' (Burt,

2005: 47). Nevertheless, he also goes to some lengths to point out that people often do not recognize their true situation. People seem not to be very good at seeing structural holes 'from the inside,' unless they already live in a hole-filled world. For example, in summarizing several studies (Freeman, 1992; Janicki and Larrick, 2005), he concluded, 'the summary conclusion from Freeman's and Janicki's experiments is that experience matters: people who live in a network that contains structural holes are more likely to recognize the holes in their next networks' (Burt, 2005: 77).

Burt's line of reasoning leads us to conclude that selection forces generated by structural ties, rather than strategic action, may account for much of an entrepreneur's structural advantage. Being able to recognize, acknowledge, or create opportunities may depend upon one's position within a social network. Having a non-homophilous entrepreneurial team, including strangers, could motivate and facilitate a search for structural holes. Random networks are full of structural holes, but most have no value and finding the few that do is a discouraging task. Truncated scale free social networks – if perceived for what they are – contain holes, but confront would-be opportunists with a daunting challenge, as we will point out shortly. Small world networks seem to be the most rewarding for those far-sighted entrepreneurs adept at building bridges between clusters.

Seeking short cuts

We can analyze effective entrepreneurial search behavior within social networks by drawing upon ideas from research on social identity. We noted the overwhelming task facing potential entrepreneurs who search for information and resources in truly random networks. A 'broadcast' search involves pursuing all possible social ties, a costly endeavor, given the lack of meaningful structure in the network. Even if we accept the implausible assertion underlying Figure 1 that we live in a world connected at random, knowing that d is only 3 is worthless to a person searching for unknown others. Because the paths to all others in the network are randomly generated, there is no *a priori* method for picking a path that will lead to any particular target. To ensure success, an entrepreneur would need to explore every single path at every single stage of d . Such a broadcast search would be incredibly costly and time consuming (Barabási, 2002).

By contrast, small worlds contain meaningful structures at the level of local clusters, and thus it would

seem possible to engage in a 'directed' search that would economize on search costs. However, we have emphasized the high degree of redundancy in local clusters, implying that valuable information lies on a path that goes beyond the boundaries of the cluster and leads to other clusters. Therefore, people searching for resources and opportunities outside of their own cluster will need to find an indirect path to them.

In a small world, 'shortcuts' to unknown others *can* be found through directed search (Watts, 2003: 245). Initially, it might seem absurd to think that anybody in Nebraska would know to whom to send a letter, when the target is a stockbroker in the Boston area (Milgram, 1967). However, people use their knowledge of *social identities* to figure out where to start. They not only know who their friends are, but also to which social categories ('identities') they belong. Thus, in deciding on how to conduct their directed search, people pick categories such as 'works in finance' or 'travels a lot to the east coast.'

These social categories enabled people to bypass the need for a random search through all of their ties. Indeed, if they had to use a broadcast search—a way to take advantage of networks that might contain a high degree of randomness—they would overwhelm the system. Rather than a broadcast search, they choose people whose social identities might move the letter closer to the target. Milgram's subjects took advantage of their knowledge of local clusters plus their implicit recognition that there might be bridges across the gap between their local cluster and the target in Boston. Knowing someone in a particular identity class or cluster is a huge first step toward a search shortcut that jumps over the otherwise random steps required to complete a search for targets lying outside of local clusters.⁹

Searching in scale free/fat tail networks

In truncated scale free networks, centrally located actors occupy positions that give them many advan-

tages over less central actors. As Sorenson and Stuart (2001) observed, central actors can 'see' more of the network and spot potential opportunities before others. Their central position enables them to mobilize collective action quickly and efficiently, such as forming entrepreneurial teams, investment syndicates, and other collective commercial activities. They can treat some actors preferentially by granting them access to information and resources over other actors, such as competitors. Indeed, because of the role played by central actors, we speculate that the likelihood of developing successful 'high-growth' ventures increases if new ventures emerge in environments displaying scale free network characteristics.¹⁰

If entrepreneurs recognize that a network relevant to them has the property of being scale free, or nearly so, what strategies should they adopt? How does someone change standing in a truncated scale free distribution? One possibility is to form ties with relevant people who are further up in the distribution. For example, in academic scholarship, victors in the struggle for citations usually work in teams, rather than going solo. Junior scholars improve their chances by establishing coauthorship relationships with senior scholars, as shown by Guimerá *et al.* (2005). Another possibility is collective action, involving coalition formation with others to take collective action. For example, almost all high-tech new ventures involve teams, rather than solo entrepreneurs or partners.

Summary

Effective search behavior in small and truncated scale free worlds depends upon someone's network location, with densely connected clusters often leading to 'echo' rather than new information. Strategic responses to the constraints of clusters require people to break out, using technological assistance as well as deliberately seeking social locations in and around structural holes. Such locations increase the likelihood of someone recognizing opportunities and encountering non-redundant information. When entrepreneurs must search beyond their direct ties, they can use their knowledge of social identities as a starting point, rather than searching randomly. Search in scale free or fat tail networks is simpler

⁹We note that using 'social identities' in searching within networks resembles social network analysts' strategy of giving survey respondents 'position generator' lists. In such surveys, respondents are asked if they know people in certain categories. For example, in the Research Triangle Park area of North Carolina, Aldrich, Elam, and Reese (1996) asked about experts in law, accounting, and finance, along with an expert in the person's specific industry. Marsden (2005: 15) noted that 'studies suggest that respondents recall alters in social clusters when answering name generators. The basis for clustering likely varies across situations, but it is possible that foci of activities such as families, neighborhoods, workplaces, or associations offer a framework for remembering others.'

¹⁰For peripheral actors, over-reliance on a central actor creates potential vulnerabilities, unless they can create additional ties to others.

than in random or small worlds, insofar as uncovering the fundamental structural order is concerned. However, latecomers to well-institutionalized fields will have to wait their turn for a place at the table or else join with others in organizing an effective vehicle for collective action.

CONCLUSIONS

Our discussion of the differences between random, small world, and truncated scale free networks has highlighted key features of each. Identifying the concepts and principles for each provides analytic clarification, but we recognize that entrepreneurship researchers will not discover distinct cases of each. Instead, we recommend that investigators try to find the conditions under which each formulation is most useful in understanding a particular entrepreneurial issue.

Our review of the three network models raises the question of **which model might be most useful** in trying to describe the contexts in which most entrepreneurs live and work. Throughout our paper, we have emphasized the importance of covering the entire range of entrepreneurial activity, beginning with attempts to found ventures and continuing through their attempts to grow. To capture the dynamics and complexity of this process, we think it is important to keep an open mind regarding which network model is most appropriate and to specify under what conditions it might arise. Because of our training as sociologists, we have been drawn to the social conditions surrounding the founding of new ventures, particularly with regard to the kinds of teams that are assembled. Although these ventures are not the 'black swans' of the entrepreneurial finance literature (Taleb, 2007), they are the raw materials from which the economic system draws much of its vitality.

We suggest that investigators wishing to use network characteristics in their models should clarify what kind of network they are positing: random, small world, or scale free. We have argued that outcomes might well be different within each model and therefore network model type may act as a moderating condition. For example, **we have hypothesized that the characteristics of entrepreneurial teams might differ across network context**, with mundane entrepreneurial ventures more likely to arise in small world contexts and high growth oriented ventures

with larger and more nationally recruited founding teams arising in scale free networks.

To conclude, we offer some questions about the implications of the three models for entrepreneurial behavior. Are **entrepreneurs in a random world, only a few steps from everyone else, but blind as to how they navigate in it?** Are they **in a very small world, sheltered and even trapped within narrow communities?** Alternatively, are they in the small world that Milgram first described, clustered to a great extent but able to take advantage of a few 'long leaps' that **connect their cluster to other clusters far removed, if only they knew how to search?** Do they become better at searching as they learn to recognize structural holes and as they take on an entrepreneurial identity?

Are they **in a truncated scale free world, where their fate depends upon where they stand in the pecking order of ties and affiliations?** Are they in a complex scale free world, trapped somewhere between order and chaos? **In such worlds, most of life is somewhat predictable, in the short run, but the significant big-impact events are rare and predictable only after the fact.** In this world, entrepreneurs are plagued with various cognitive heuristics and biases, and rewards come in the form of very lumpy payoffs (the occasional 'black swan'). Which of these worlds did Simmel have in mind when he wrote of opportunities proliferating into infinity?

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